Say we want to find general solution to:

$$a(t)y'' + b(t)y' + c(t)y = f(t)$$

Still linear and usually nonhomogeneous but doesn't have constant coefficients is big difference.

And that is a big change as we haven't really even touched the idea of solving second order equations with nonconstant coefficients prior to now.

Turns out variation of parameters can actually help us solve such equations, at least sometimes. Ok still very delicate, but fact that it can do that is quite remarkable as that seems so different.

Shows the importance of it as undetermined coefficients won't really even be a realistic option in such situations.

But how?

Say y_1, y_2 are the two linearly independent solutions to the complementary equation, that is, solutions to:

$$a(t)y'' + b(t)y' + c(t)y = 0$$

Let us still try for a particular solution:

$$y_p = u_1(t)y_1(t) + u_2(t)y_2(t)$$

Compute y_p' then again assume $u_1' y_1 + u_2' y_2 = 0$

Same calculation then (did this last time). Turns out $a(t), b(t), c(t)$ being nonconstant doesn't affect at all how this simplifies, we will essentially get same thing:

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{f(t)}{a(t)} & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & \frac{f(t)}{a(t)} \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

Ok, sure $a(t)$ is function here, not constant but still fundamentally same looking formula.

So since this seems to work just as well with nonconstant coefficients as constant, why didn't we just do this from the get go? Seems like it would have saved us time.

Most people's first instinct is because having $a(t)$ in formula instead of constant a is reason, that maybe that makes the integration much harder. But that's not really why.

Sure $a(t)$ being function not constant will change things, but integrals can be deceptive, having function instead could actually make the integrals needed easier in some situations! Very situation dependent in that respect.

The real problem with this is to find y_p you need y_1, y_2, the solutions to the complementary equation.

But how to find those? In constant coefficients we used characteristic equation, that's not an option with nonconstant coefficients.

Whether there is a convenient way to find y_1, y_2 like many things depends on the problem, but certainly not something we've really done.

So really perhaps in grand scheme of things, finding homogeneous solution is harder than moving from homogeneous to nonhomogeneous.

B4.6.23

$$ty'' - (t + 1)y' + y = t^2$$

We are given homogeneous solutions $y_1 = e^t$, $y_2 = t + 1$ and want to find a particular solution.

So we try:

$$y_p = u_1(t) \cdot e^t + u_2(t) \cdot (t + 1)$$

Then by usual formulas listed above:

$$u_1' = \frac{\begin{vmatrix} 0 & t+1 \\ t & 1 \end{vmatrix}}{\begin{vmatrix} e^t & t+1 \\ e^t & 1 \end{vmatrix}} = \frac{-t(t+1)}{-te^t} = (t+1)e^{-t}$$

---> $$u_1 = \int (t+1)e^{-t}\,dt$$

by parts

$$= -(t+1)e^{-t} + \int e^{-t}\,dt$$

$$= -(t+2)e^{-t}$$

and then:

$$u'_2 = \frac{\begin{vmatrix} e^t & 0 \\ e^t & t \end{vmatrix}}{-te^t} = \frac{te^t}{-te^t} = -1$$

---> $$u_2 = -t$$

Hence general solution is:

$$y(t) = y_c + y_p$$

$$= c_1 e^t + c_2 (t + 1) - (t + 2) - t(t + 1)$$

$$= c_1 e^t + c_2 (t + 1) - (t^2 + 2t + 2)$$

So how would we find y_1, y_2 if not given? Well as we did in Calculus, find series solution maybe, i.e. Taylor/Maclaurin Series.

But that's rather dicey, even if you do that, what is the guarantee that it will simplify to some simple closed form to enable you to do this as we did in this problem? No real assurance of that unfortunately in general.

Chapter B9

In this chapter our main aim will be to solve "systems" of differential equations. And since we are solving systems, we might use matrix ideas to help us.

This chapter is really the epitome of why we do both linear algebra and differential equations in this class, as mentioned we will be solving certain differential equations with the help of linear algebra ideas.

We did do a bit of systems of differential equations in B4.6, variation of parameters, but honestly the ideas in this chapter are fairly different from that setting.

Also note, the book we are using now is inherently separate from the linear algebra book. As such they do not assume you know those crossover ideas. Hence we are skipping sections B9.2 - B9.4 since the ideas there we mostly already covered (e.g. eigenvalues).

We will also talk about some physical applications of these systems, which we mostly skipped in Chapter B4 since those weren't really new to us in theory (went over them in Calculus).

B9.1: Notation

We will focus on writing the equations we wish to solve in this section, that is the proper notation for them, again usually using matrices. The actual solving will be in B9.5 - B9.6

Say x_1, x_2 are functions of same variable t.
Then a potential system of differential equations for these are:

$$x'_1 = -4x_1 + x_2$$
$$x'_2 = 3x_1 - 2x_2$$

So in short, if we had x_1, x_2, differentiate both and plug in, both equations should be satisfied.

Where might we have seen system of differential equations in Calculus? Most common place would have been predator-prey (or similar models are competition, cooperation).

Say x_1 is prey and x_2 is predator. Potential system is:

$$x'_1 = 2x_1 - 5x_1x_2$$
$$x'_2 = -3x_2 + 4x_1x_2$$

Prey increases on its own hence the $2x_1$ term, but decreases due to interactions with the predator, hence the $-5x_1x_2$ term. Similarly predator decreases on its own while increasing due to interactions with the prey.

This is very natural context for system as each's population certainly affects how the other one changes.

But unfortunately solving this is not really realistic for us like the prior

$$x_1' = -4x_1 + x_2$$
$$x_2' = 3x_1 - 2x_2$$

as due to the $x_1 x_2$ terms, the predator-prey model is not linear like the above (after all $x_1 x_2$ is a quadratic style term). Naturally if we are going to use matrix style ideas, we do probably need linear equations as we had earlier in course.

One natural reason is that in the linear system feels like both being of style Ce^{kt} might work as all terms then of that style so makes sense it could work. With $x_1 x_2$ term that won't work as that will create e^{2kt} which won't be in sync with the other terms.

So another context we did in Calculus which, when expanded, might give us a linear system is <u>mixing problems</u>.

Usual context was a tank with liquid that has some salt mixed in is drained / replenished. And depending on circumstances, the amount of salt in the tank likely changes. So our dilemma was to find a formula for the amount of salt in tank at time t. Perhaps we need to make sure amount doesn't drop too low so that if there are fish in tank, they don't survive.

But ok, that wasn't a system, just a single differential equation. Suppose though we expand on this.

Say we have two tanks which are "feeding" each other, each sending some of its contents to the other, and perhaps one or both are being drained/replenished externally.

For instance in

$$x'_1 = -4x_1 + x_2$$
$$x'_2 = 3x_1 - 2x_2$$

the $-4x_1$ in first equation could represent the salt flowing out of first tank while the $+x_2$ could represent the salt flowing in from the other tank. And similar for other equation.

Can write this with matrices:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} -4 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

What about solving this? Comes a bit later. But good to note first that when we were solving matrix equation $Ax = b$ with A square, we would often left multiply by A^{-1} to get $x = A^{-1}b$

Might we try to solve above system like that, using inverse? Unfortunately here the "b" is not constant, it has the variables in it (their derivatives in fact) so that won't really solve it.

So despite the similar looking matrix notation the ideas needed to solve are quite different. But point is the setup and ideas will work same whether matrix is 2 x 2 or 10 x 10. Sure calculation is inherently longer with larger matrix as in any setting but again ideas will be same.

Can even write single linear differential equations (of any order) in this form. Say we have:

$$ty''' + t^2 y'' + y' + 5y = 0$$

Think of $x_1 = y, x_2 = y', x_3 = y''$

and seeing that the current equation can be written as:

$$y''' = \frac{5}{t}y - \frac{1}{t}y' - ty''$$

overall this differential equation can be expressed as:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{-5}{t} & \frac{-1}{t} & -t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Unlike prior, not just constants here.

Important to realize, I'm not claiming this makes the equation easier to solve, just useful to have another way to look at it.

For instance a linear homogeneous second order equation which has constant coefficients, that is

$$ay'' + by' + cy = 0$$

thinking of $x_1 = y$, $x_2 = y'$, by same idea we can write this as:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ \frac{-c}{a} & \frac{-b}{a} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

and here this matrix does have only constants so more accessible to solve probably using the upcoming ideas, this is what we largely did in variation of parameters.

Now we try to move to solving.

B9.5 Solving systems of differential equations

For set of n functions, $x_1, x_2, \ldots, x_n$ of same variable t, writing

$$X = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

a linear system of differential equations is:

$$\underbrace{x'(t)}_{n \times 1} = \underbrace{A(t)}_{n \times n} \underbrace{x(t)}_{n \times 1} + \underbrace{f(t)}_{n \times 1}$$

where $A(t)$, $f(t)$ contain functions of t (no restriction on which functions).

Using some old terminology/terminology from B9.4 (which we are skipping but using some notation from there), we say

(1) When $f(t) = 0$, this is called <u>homogeneous</u>.
(2) When $A(t)$ has only constant entries, this is called <u>constant coefficients</u>.

We will mostly solve systems where (1) and (2) both hold. Might dabble a bit into nonhomogeneous, but constant coefficients is pretty forced, just not generally accessible to deal with nonconstant coefficients.

For instance the earlier system

$$x'_1 = -4x_1 + x_2$$
$$x'_2 = 3x_1 - 2x_2$$

which has constant coefficients, we speculated this could represent a mixing problem with two tanks.

Makes sense that it would have constant coefficients as the flow rates presumably stay the same.

But what if one / both are being replenished externally rather than just from other tank? That might turn that system from homogeneous to nonhomogeneous.

Hence in theory why we might want to be ready for both.

Or what if say the amount of liquid in one / both tanks wasn't constant. Then amount of salt per unit unit of liquid won't stay in consistent in ratio to the amount of actual salt so this probably won't be constant coefficients anymore.

Ok, our ability to solve those problems may not be high (particularly the latter type), but it's interesting to think about those nuances that we came across when dealing with a single tank in Calculus (and there we could generally deal with those nuances even if they made things harder) play into the differences we get in systems. I.e. the same types of things that concerned us in the last chapter, homogeneous vs nonhomogeneous and constant coefficients vs. nonconstant coefficients.

Will go into actual solving next time.